

Year 1 Electromagnetic Calorimeter Reconstruction and Calibrations

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1 Introduction

2 Raw Data Processing and QA

2.1 Signal extraction

During the 2023 commissioning period, 31 samples were recorded for each EMCal tower per an event. The calorimeter was timed in such that the peaks of signals associated with particle energy deposits were around the 7th sample (commonly referred to as sample 6). Every tower was processed by the CaloWaveformProcessing module in the TEMPLATE mode. This mode fits a “template” waveform to calorimeter signals. An illustration of a fitted pulse can be seen in Fig. 1. The template, before fitting, has its first few points at zero (the pedestal) and the highest part, the peak, at the y -axis value of one (the scale or peak minus pedestal) and a x -axis value of approximately six (the time). The free parameters of the fit are the pedestal, peak time, and scale. The CaloWaveformProcessing reports these parameters for every tower fit, along with a goodness-of-fit metric termed χ^2 which is the summed squared deviation of the samples to the fitted template values, all divided by the number of samples fit.

There are two major pathologies relating to the tower signals recorded and template fitting of these signals. First, are noisy bits in particular ADC boards/channels. This manifests itself in a single sample in a waveform being unnaturally large. These “bit flips” range in their rarity for the small number of problematic channels. Because the single sample is large, it can lead to a large extracted signal amplitude by the template fit but also results in a concomitant large χ^2 , because the large sample is not accompanied by the signal shape in the surrounding channels, unlike the good pulse seen in Fig. 1. The second is the improper configuration of a few ADC channels, where the boundaries of 14-bit words that form individual samples are not correct, leading to a shifting of bits and a wrap-around-like effect. This manifests itself as all samples in every event being multiplied by a factor of 2^N where N is usually 7 and a wrap around of a pulse instead of the pulse just saturating at 2^{14} . This leads to larger signals and larger noise for the channel and a larger χ^2 . Towers that exhibit this pathology are termed “bit shift” towers.

These two pathologies cause distortions in the extracted energies that would be too time consuming, relative to the time invested in fixing the underlying issue, and thus are simply removed from the data analyzed in the commissioning period. This removal is achieved with the χ^2 . A correlation of the tower energy (with an approximate calibration) correlated with χ^2 for all tower for all events in the run noted in the plot. The horizontal correlations at approximately 5e3 and 5e5

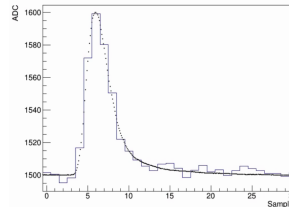


Figure 1: An illustration of the template fitting procedure with the calorimeter signal shown as a histogram and the fitted template shown as the dotted distribution.

are associated with bit-shift towers. The figure demonstrates that these bit-shift towers regularly produce very large extracted energies and are termed “hot”. To isolate towers and events of particular towers, an extracted χ^2 of greater than $1e4$ was chosen to be a bad χ^2 value. This is flagged in the tower objects (TowerInfov2) via the `get_isBadChi2()` member function returning true. All towers with a fraction of hits (approximately greater than 0.5 GeV) with a bad χ^2 that is over 1% for a given run are masked for the entirety of that run. This is flagged by the `get_isHot()` member function returning true. Also note that the single sample noise of the EMCal towers ranges from 5 – 15 ADC and thus one expects normal χ^2 values to be around 25 – 225.

A recommended set of timing cuts to reduce noise is also offered. The mean time for hits for each channel are shown in the right panel of Fig. 2. If an event tower’s extracted time is not within ± 1.0 samples of the mean that tower in that event is flagged by `get_isBadTime()` returning true.

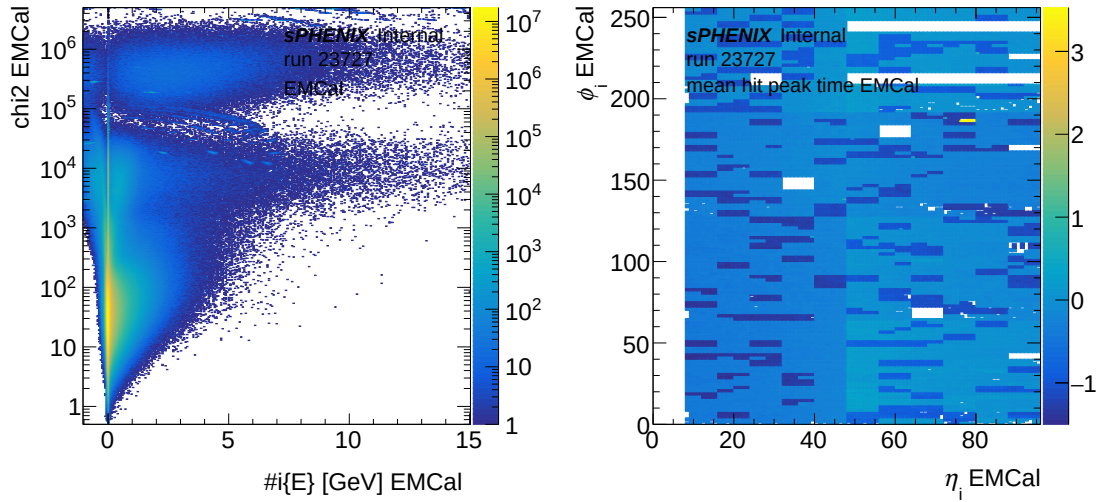


Figure 2: Left: correlation between energy and template fit χ^2 and calibrated energy for all EMCal channels. Right: the mean peak time (relative to sample 6) of individual channels for towers above 150 ADC.

2.1.1 Noisy Tower Removal

Noisy towers are flagged in the EMCal after signal processing. The frequency of hits above a given threshold is measured for all the towers in the EMCal for a given run, and the width of the result distribution is then used to calculate cut-off values which determine which towers are to be masked. For this analysis in particular, a low threshold value of 15 ADC is used, which results in an average frequency of about 25% is found, and a wide cut of 2.5σ is used to flag bad towers. Noisy towers are those which fire too frequently and cold towers which fire too infrequently. The following figures show an example of the flagging procedure and its impact on run 23727, which will be used as an example run throughout this document.

Figure 3 shows the fraction of events in which a given tower sees a hit above 15 ADC as a function of tower number. In this analysis, this histogram is essentially filled once per segment. Otherwise, one would have to run over the entire run linearly, which will become very time-consuming for

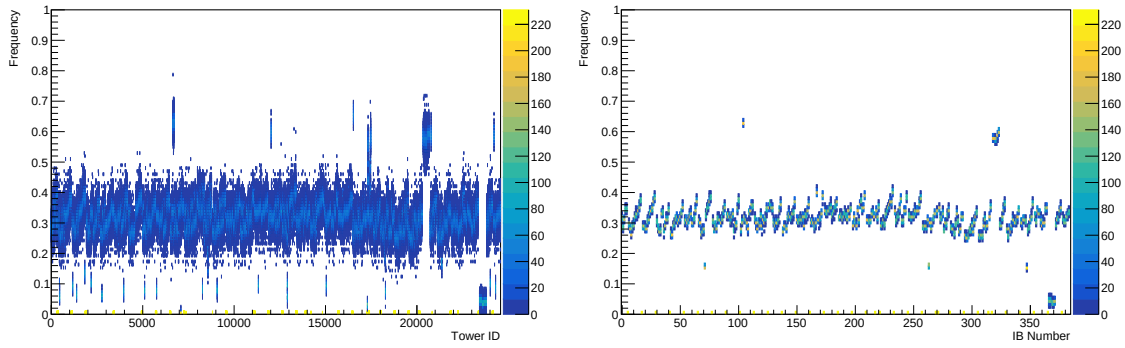


Figure 3: Left: Percentage of events in which a given tower records a hit above 15ADC as a function of Tower ID. Right: The same quantity, but this time recorded at the IB level.

larger data volumes. One can see on average, towers see a hit above 15 ADC in about 30% of the runs. A similar distribution is made for interface boards, or IB's, on the right side of Fig. 3. Because errant behavior can often be shared across an entire interface board (or even sector), the decision was made to remove entire IBs if their performance was too dissimilar from the mean. One can see though, that usually if an IB is going to get masked for some reason, all the towers individually will themselves have been flagged.

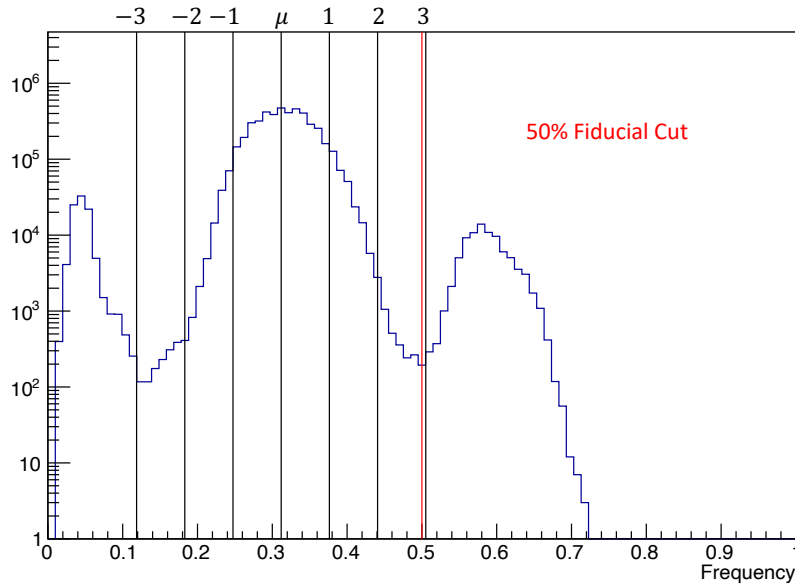


Figure 4: 1 dimensional projection onto the frequency axis of Fig. 3 for towers. The line denoted μ represents the histogram mode, and each of the black lines to the right and left of it are n sigma cutoff thresholds, where n is the number above the line. The red line at 50% is the fiducial cutoff, above which towers and IBs are automatically flagged as bad.

As stated previously, a 2.5σ cutoff is used in this QA analysis, and the impact on the detector behavior is shown in figure

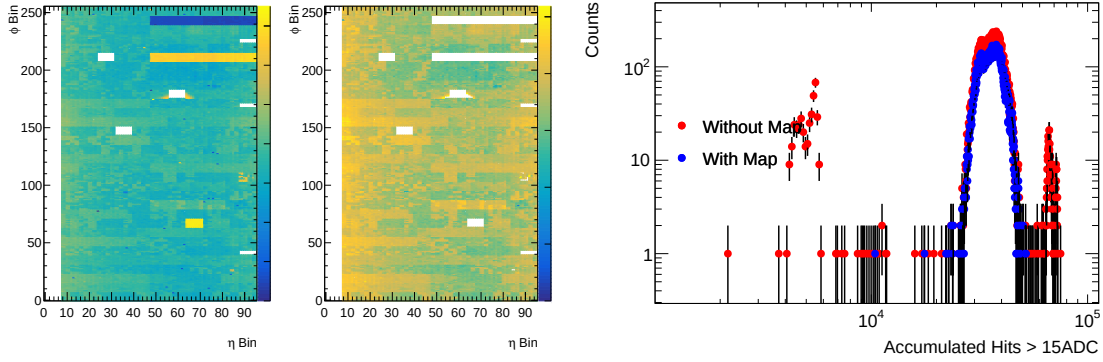


Figure 5: Left: a 2D distribution of tower hits above 15 ADC before any tower masking. Center: The same distribution after masking has been applied. Right: Comparison of 1D distribution made from the previous two plots. The red distribution is unmasked, the blue has problematic regions masked.

Figure 5 shows the impact the masking has on the EMCal's behavior in Run 23727. The plot on the far left shows the distribution of hits above 15-ADC in η and ϕ . One can see several pathologies. In the top right, a sector has gone cold, thus appearing dark blue. Two sectors down in ϕ from there, one can see a sector that is in high-gain mode, registering hits in many many more events. And finally, one can look towards an $(\eta, \phi) \approx (65, 55)$ and see a hot interface board. The areas that are white have identically zero counts and thus are either uninstrumented (as is the case with towers in η bin less than 8) or not receiving any signal due to malfunctioning hardware. The middle panel shows the impact of the masking procedure, whereby these problematic, but responsive, areas have been completely removed. The rightmost panel shows a 1D projection of the the previous plots, with the red distribution being from the unmasked data, and the blue coming from after the data has been masked. One can see the maps are removing outlier towers, leaving only the central distribution.

3 Calibration

In this section we discuss the calibration of the EMCal to the EM scale with a combination of tower energy spectra, reconstructed π^0 mass, and pre-installation cosmic ray MIP distributions. The pre-installation cosmic ray data was compared to MC simulation to set a tower-by-tower absolute scale but was not done to significant precision to make quantitative statements of beam data to MC agreement once installed. This calibration only served as an input value for beam data calibration. The cosmics are not documented here.

The beam-data-based 2023 calibration utilized a method to establish an absolute tower-by-tower calibration which is achievable with a few million events. The calibration was factorized into two procedures. The first is called the "tower-slope" calibration (TSC), which assumes ϕ symmetry in the energy deposits in towers. These energy spectra are compared between towers with the same η index to derive a relative calibration between all towers in a particular η index/slice. The second procedure constructs di-cluster mass distributions, where each distribution is all pairs with a leading cluster's leading tower determining the particular pair's η index. These mass distributions were constructed in the range of the π^0 mass which is extracted by a fit. Thus, the

shifting of the mass to a specific value (predicted by the MC) achieves an absolute calibration for the η bin, which had all the towers in this slice are already calibrated relative to one another in the first TSC. Finally, post-calibration hot tower removal is required to remove any last hot towers which may have been introduced by failed calibration procedures. The following subsections describe these steps in detail and give example results.

The data were split into groups of runs which had similar EMCal acceptance, were recorded at similar times, meet the minimum bias event selection criteria ([here](#)), and were recorded with the same trigger. The run groups had to contain at least one million events, if not, the runs were not calibrated. The two procedures outlined above were carried out with all events in the run group and resulted in a single calibration factor per a tower for all events in the run group. These calibrations were uploaded to the Calibrations Database and applied in the CaloTowerCalib module.

3.1 Relative Scale calibration (Tower Slope)

The TSC is a good way to achieve a simple relative calibration with a limited statistical sample. It makes the assumption of ϕ symmetry of all the towers, The procedure fits a particular towers energy spectra with a reference spectra $f(E)$ with a free gain parameter (g) and a scale parameter (A); thus the fit function is written $Af(gE)$. In this case the reference spectra is the ϕ -integrated spectra.

- The tower E distribution was fit between 0.2 and 0.8 GeV.

The assumption of ϕ symmetry is not strictly true. These geometric effects could possibly be corrected for with an MC-based correction but is currently not. The correction of this is made as a systematic variation.

A tower with a very poor pre-installation calibration factor or an acute hardware or readout issue can bias the reference spectra. Thus hot tower masking at the raw level was necessary (documented in the previous section). This also motivated running the tower-slope procedure twice (iteratively) to first calibrate out most of the issues and then repeat (applying the constants and re-filling the histograms) with a more appropriate reference spectra. This also provides a convenient check of the convergence of the method.

The results of this calibration procedure for one run group is shown in this subsection. The TSC factors from the first iteration are shown in the left panel of Fig. 6. There does seem to be some structure to the coefficients but they largely scatter around randomly. The TSC coefficients for the second iteration are shown in the right panel of Fig. 6. The scatter has decreased substantially as expected (note the z-axis scale). Also, it appears less “stripy” as the reference spectra are not getting distorted by a single (or a few) towers leading to most of the other towers corrections being above or below one.

The tower spectra after the TSC can be seen in Fig. 7 and 8. This shows the results after the second iteration coefficients have been applied and the histograms remade. Each η slice is shown with a similar normalization and the bin at 0.2 GeV is the same value. The reference (η -integrated)spectra are shown in red. Spectra that fail the post-TSC QA are shown in blue (discused below). This

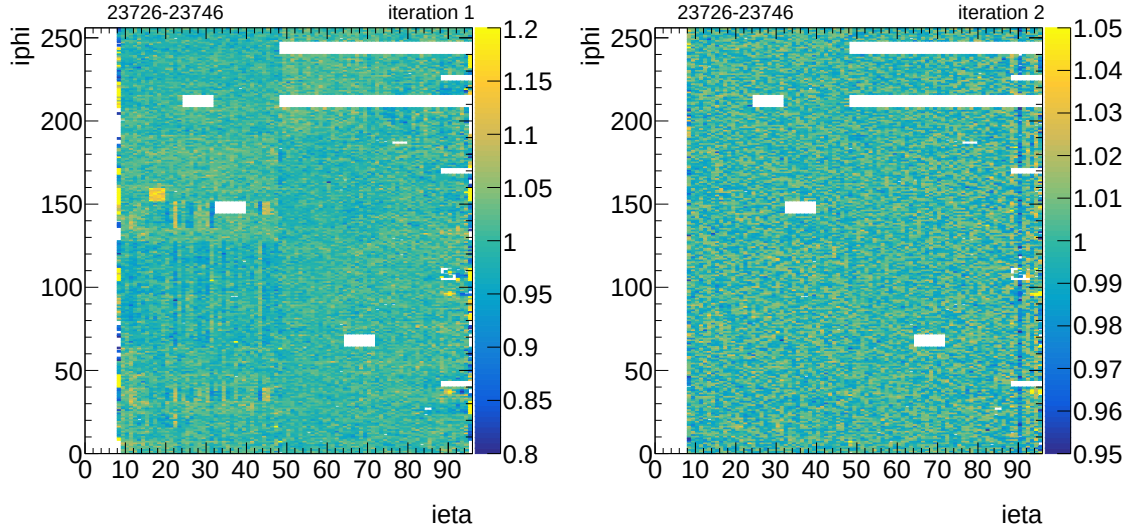


Figure 6: The calibration coefficients extracted by the first iteration of the TSC for run group 1. Left: the coefficients generated in the first iteration of the TSC. Right: coefficients from the second iteration.

demonstrates that the reference and the spectra agree quite well in the region in which they were fit. There are some spectra which seem to have larger noise but then begin to agree well once the signal dominated region is reached. There are a small number of spectra which have significantly different trends than the reference and are generally flagged by the post-TSC QA.

The post-TSC QA identifies towers which have a $5 \times \text{RMS}$ difference in the number of hits (events above 1.0 GeV). Each η slice is considered individually. The mean and RMS of the number of hits in a tower distribution for the η slice is calculated. Towers outside of the mean $\pm 3 \times \text{RMS}$ are removed and the mean and RMS are recalculated. Then towers outside the mean $\pm 5 \times \text{RMS}$ are defined as hot and flagged in the reconstruction by `get_isNoCalib` returning true. There are towers with very inordinate amount of hits compared to their η -slice neighbors and thus this simple outlier rejection ensures similar treatment of the slices, if or if no very hot tower exists in the slice.

3.2 Absolute Energy Scale (η -by- η π^0) Calibration

In the previous section towers of the same η slice were calibrated relative to one another. In this section we discuss the absolute calibration of each η slice by reconstructing π^0 mesons and measuring the mass and then adjusting the calibration such that the proper (target) π^0 mass is obtained. The observed π^0 mass is sensitive to non-calibration related effects such as detector resolution and thus a target mass that is not the PDG mass is necessary. A discussion of this target mass is located in Section 3.2.1. The procedure runs over all the data in a run group, constructing clusters with towers that have a calibration from the previous step (TSC or prior iteration of π^0 calibration). Then, di-cluster mass distributions are constructed with all kinematic selections based on the `e_core` cluster energy value. These selections are

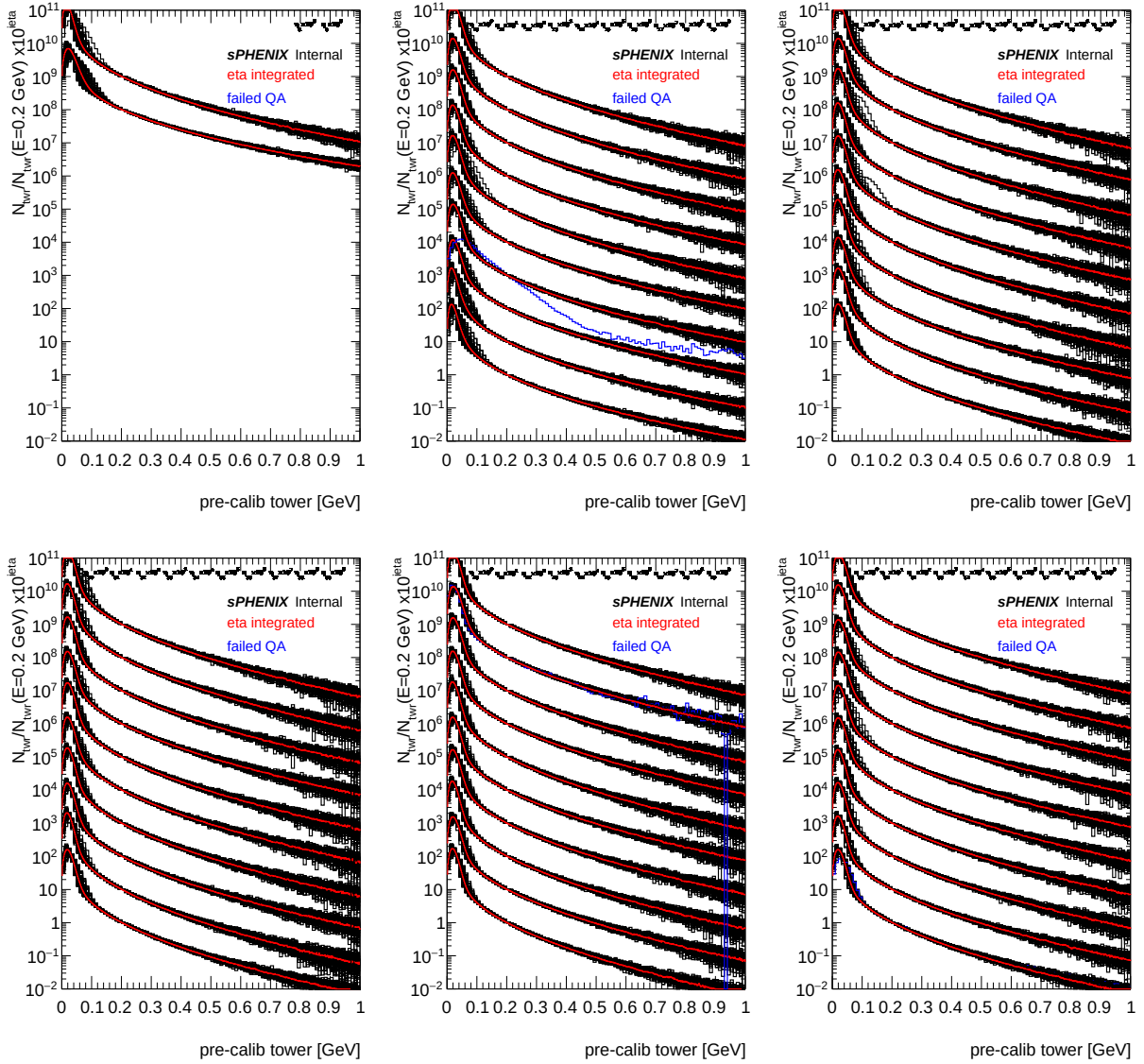


Figure 7: Tower energy spectrum for run group 1. The black distributions are individual tower spectra, which are shown in blue if the tower fails the post-calibration QA. The η -integrated spectra used as the reference distribution are shown in red. All spectra are normalized as indicated by the y-axis.

- $E_{T1} > 1.3 + 1.4 \times N_{clus}/200$ for $N_{clus} > 30$ and $E_{T1} > 1.4$ for $N_{clus} \leq 30$
- $E_{T2} > 0.7 + 1.4 \times N_{clus}/200$ for $N_{clus} > 30$ and $E_{T1} > 0.7$ for $N_{clus} \leq 30$
- $p_T^\pi > 1.2(E_{T1}^{cut} + E_{T2}^{cut})$
- Clusters must have a cluster $\chi^2 < 4$
- $\Delta R_{\gamma\gamma} < 1.1$
- where N_{clus} is the number clusters with $p_T > 0.5\text{GeV}$ and $\chi^2 < 4$

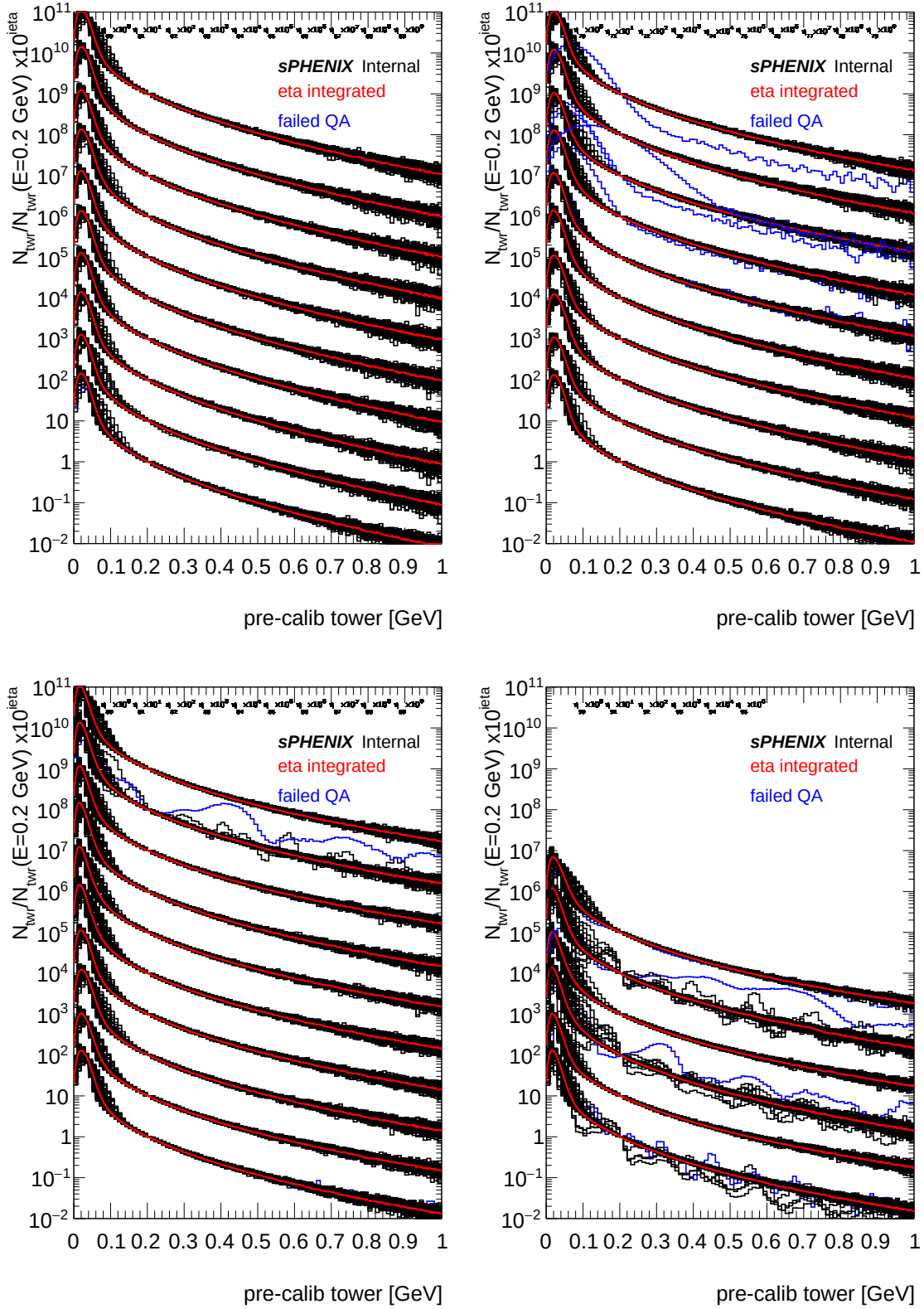


Figure 8: Tower energy spectrum for run group 1. The black distributions are individual tower spectra, which are shown in blue if the tower fails the post-calibration QA. The η -integrated spectra used as the reference distribution are shown in red. All spectra are normalized as indicated by the y -axis.

The pairs of clusters are organized into histograms to best characterize the calibration of one η slice. This is performed by filling the pair in the η slice/index of the cluster 1's (E_{T1}) leading tower index.

The mass histograms are self normalized and fit with

$$[0]/[2]/2.5\exp(-0.5((x - [1])/[2])^2) + [3] + [4]x + [5]x^2 + [6]x^3 \quad (1)$$

a gaussian in combination with a 3rd order polynomial. Some parameter limits are implemented to ensure convergence as well as seed parameters from a gaussian fit to the peak region. The parameter [1], representing the π^0 mass, is compared to the target mass value and all the towers in that η slice's calibration are multiplied by the ratio of the target mass over the observed mass. Then the procedure is re-ran. Many iterations (20-30) are run to achieve sub-statistical fluctuations in the observed π^0 mass.

A set of results from the fits are shown here. This is after the last iteration of the method. All the mass spectra with the fit is shown in Fig. 9. The fits describe the data well. The fit parameters are shown in Fig. 10 (for the π^0 gaussian part) and Fig. 11 (for the background). One can see that shifting of the mass via the updating the calibration is working and all the reconstructed masses are centered around the target mass.

The final calibration coefficients for the run group are shown in Fig. 12. It show that the calorimeter has an approximate scale of 3 MeV per an ADC.

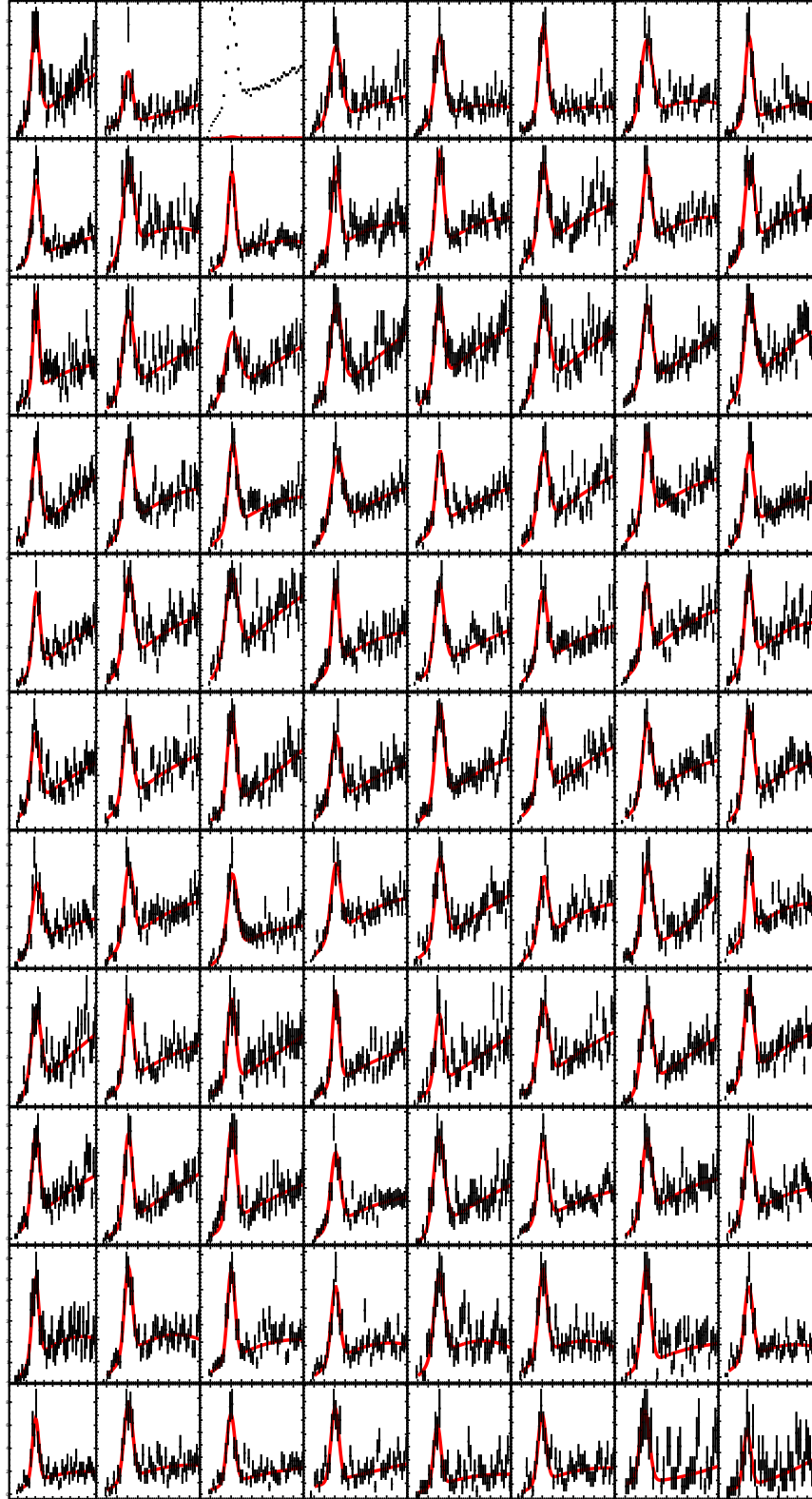


Figure 9: All the di-cluster mass spectra of the last iteration of the π^0 calibration for run group 1. The data is shown in black markers and the fit is shown in red lines

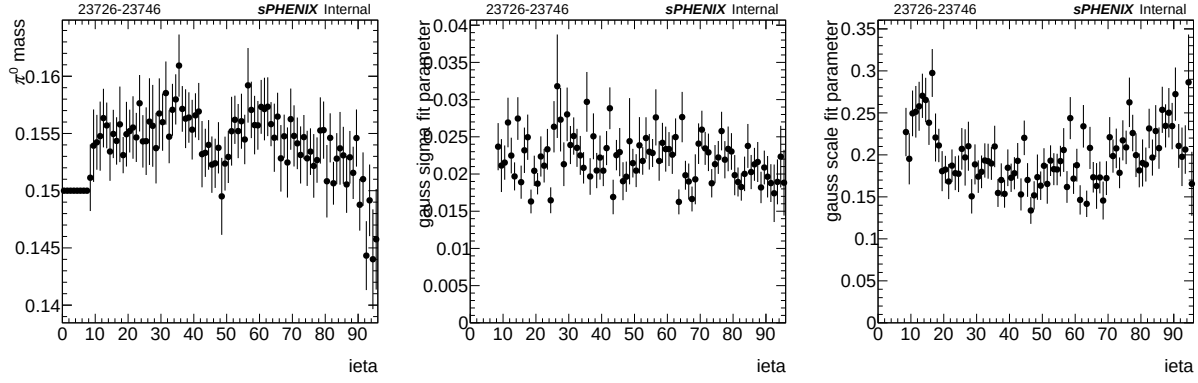


Figure 10: The π^0 fitted parameters of the last iteration of the π^0 calibration.

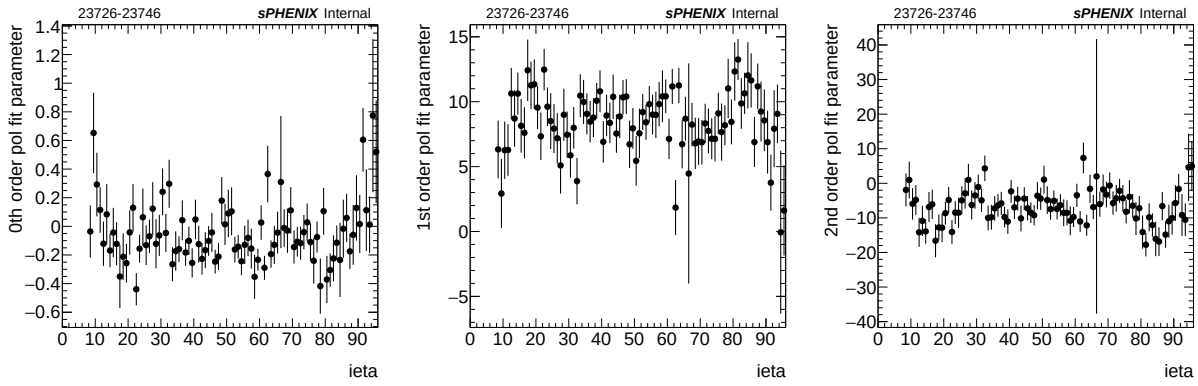


Figure 11: The di-cluster background fitted parameters of the last iteration of the π^0 calibration.

3.2.1 Selection of Target Mass for π^0 Calibration

There are many effects, beyond calibration, that effect the observed π^0 mass. These include,

- momentum spectra of the π^0
- detector energy resolution
- occupancy
- position resolution.

It is assumed that the sPHENIX Geant simulation and the HIJING MC generator adequately simulates all of these factors. However, because this first calibration has limitations the energy resolution will be worse in data. In the following two subsections we discuss the nominal prediction from HIJING and the measured resolution and the resultant adjustments to the target mass.

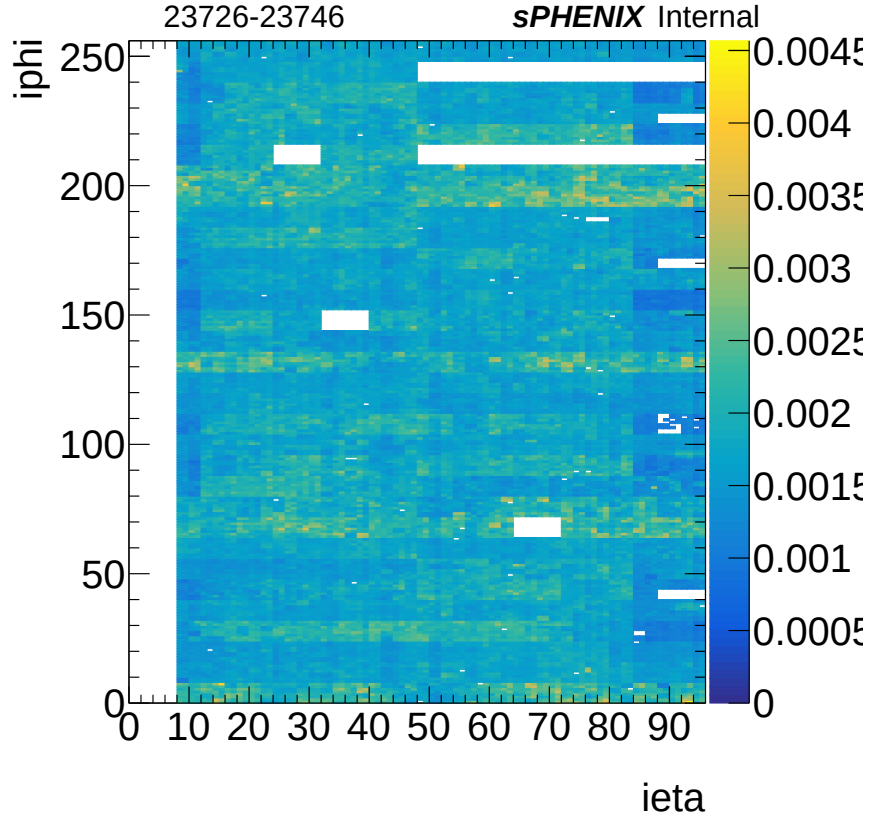


Figure 12: The final EMCal calibration coefficients for the noted run group.

206 3.2.2 HIJING predicted π^0 mass

207 In this section we discuss the observed π^0 mass in run10 minimum-bias HIJING. The same
 208 di-cluster selection criteria was imposed as the data, noted in Section 3.2. The observed mass as a
 209 function of leading-tower η index is shown in Fig. 13. There are many effects which determine
 210 the shape of this distribution. First, a single calibration coefficient (sampling fraction) is used
 211 for every tower in the sPHENIX EMCal for MC. Towers are organized into four-packs of tilted
 212 blocks, thus every other tower has a block boundary on either the edge facing $\eta = 0$ or $\eta = \text{inf}$
 213 giving an oscillatory response which is not accounted for in the cluster reconstruction. The second
 214 geometric effect is the non-productivity of the central three blocks. The remainder of the effects
 215 are probably due to physical effects. First, the resolution at the fixed E_T threshold is improving at
 216 forward η . This can have the effect of lowering the mass. Second, due to the forward/backward
 217 asymmetry of sPHENIX (tracker readout electronics and beam pipe flange) a greater “underlying
 218 event” exists in the $\eta < 0$.

219 3.2.3 Target mass adjustment due to data-MC differences

220 To further study the target mass from HIJING and MC in general, we conducted a single particle
 221 MC (SPMC) simulation of neutral pions in Geant4. Pions are generated with a flat p_T spectrum

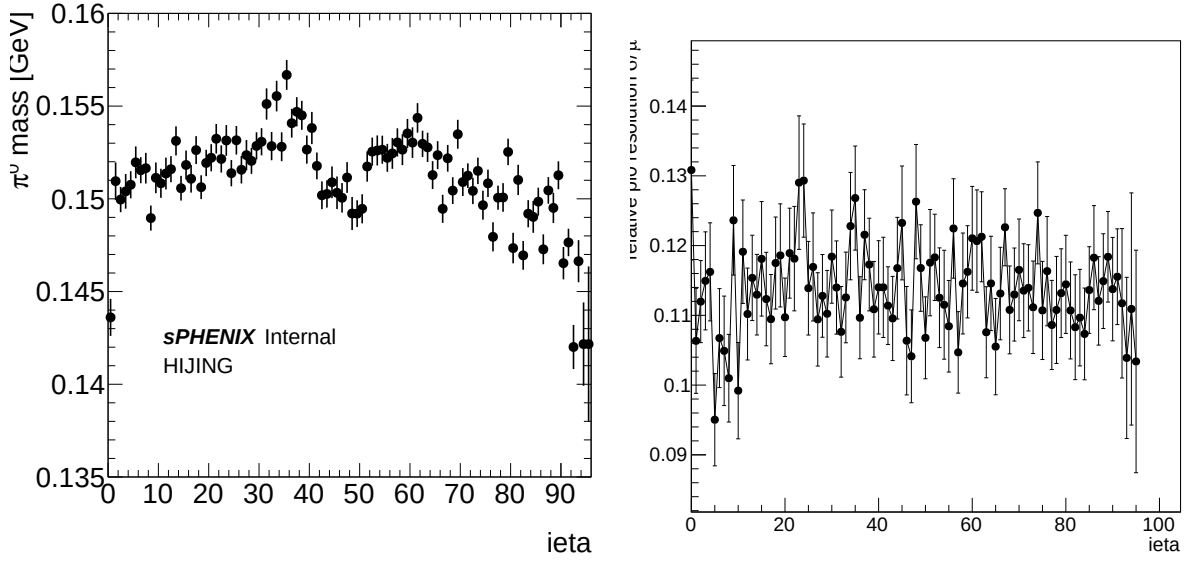


Figure 13: Left: The observed π^0 mass as a function of leading tower η index in the Hijing MC generator. The statistical uncertainties are not plotted here. Right: the relative resolution of the π^0 mass peak in HIJING.

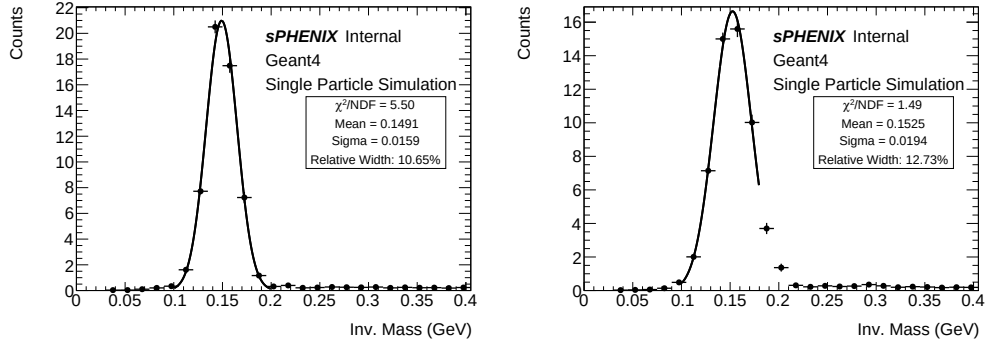


Figure 14: Neutral pion (π^0) Invariant Mass simulation (Geant4) with no additional smearing (left), and with additional smearing (right).

between $[0.2, 10]$ GeV. The pions are then weighted with a parameterization of the pion p_T spectrum observed by PHENIX. The parameterization is as follows,

$$\frac{d\sigma}{p_T dp_T} = T(p_T) \frac{A}{(1 + p_T/p_0)^m} + (1 - T(p_T)) \frac{B}{p_T^n}, \text{ where } T(p_T) = \frac{1}{1 + \exp(\frac{p_T - t}{w})} \quad (2)$$

and $t = 4.5$, $w = 0.114$, $A = 229.6$, $B = 14.43$, $n = 8.1028$, $m = 10.654$, and $p_0 = 1.466$ [1, 2]. In order to facilitate a direct comparison between data and the SPMC, the same cluster cuts were used in both. The cluster cuts are those in 3.2 with the modification that all N_{clus} -dependent p_T cuts were removed. In addition, in the data, an $N_{clus} < 75$ cut was applied to limit occupancy effects which, by construction, are not present in the SPMC.

The relative resolution (σ/μ of a gaussian fit to the peak) observed in the SPMC is 10.6%. The

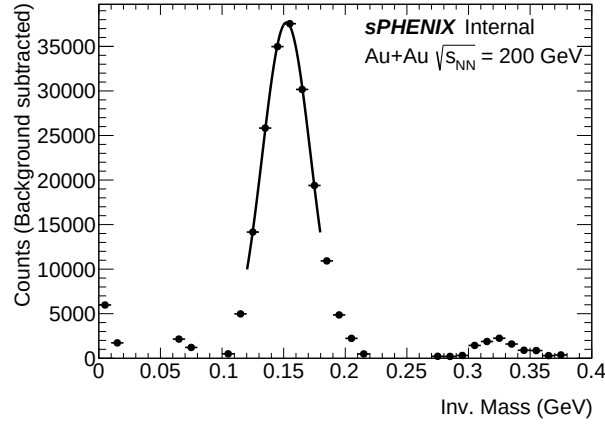


Figure 15: Neutral pion (π^0) Invariant Mass from run 23 data where the background is subtracted.

observed resolution in data is 12.86% (seen in Fig. 15, the background subtracted mass distribution). To address this data-MC difference an additional smearing was added to the SPMC. This involved scaling the cluster four-vectors by a factor,

$$F = 1 + \text{Gaussian}(\mu = 0, \sigma) \quad (3)$$

With $\sigma=12.5\%$, the measured relative resolution increased to 12.7%, which matches data within uncertainties. These peaks are shown with their fits in the right panel of Fig. 14. The additional smearing resulted in an expected shift of the peak position in the SPMC, from 149 to 152 MeV.

Smearing Added	Extracted Mass	Relative Width
0 %	148.6 MeV	9.8 %
11.5 %	152.0 MeV	12.5 %
12.5 %	152.5 MeV	12.8 %
13.5 %	153.2 MeV	13.1 %

Table 1: Relative width and invariant mass peak position for various levels of smearing added to the SPMC

A correction Factor for the HIJING mass peak can be taken from these simulations, as the ratio of the peak positions of the smeared SPMC to the SPMC with no additional smearing. This factor is found to be $152/149 \rightarrow 102\%$.

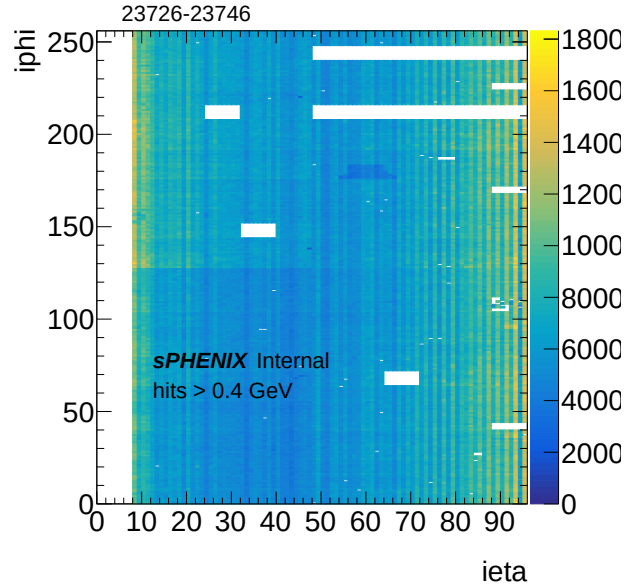


Figure 16: An η - ϕ map of accumulated tower energies with a 0.4 GeV energy cut.

4 Validation

One simple validation is the number of hits above a threshold as a function of $\eta - \phi$. This is very sensitive to hot towers. This can be seen for the run group in Fig. 16. As one can see there is no single tower dominating the scale and we see the oscillatory behavior in tower η index discussed in Section 3.2.1.

5 Recommended/Necessary User QA Cuts

The calorimeter calibrations group recommends that EMCal-based analyses only be performed with the calibrated runs documented in Section 6. There are four tower statuses to note: `isHot`, `isBadChi2`, `isNoCalib`, and `isBadTime`.

- `isHot` is set to `true` for specific towers for an entire run and should always be applied for physics analysis.
- `isNoCalib` is set to `true` for specific towers for an entire run groups and should always be applied for physics analysis.
- `isBadChi2` is set to `true` on an event-by-event basis and should be used for physics analysis.
- `isBadTime` is set to `true` on an event-by-event basis and is of unproven utility at this moment but tends to only effect very low energy signals.

A user-friendly implementation of the statuses is included, where if all the previous mentioned statuses are set to `false` the member function `get_isGood` will return `true` and is recommended for

257 physics analysis.

258 6 Calibration Run List

259 As discussed in the introduction of Section 3, the data is split into blacks of runs which are
260 treated as one and all the calibration factors are the same. The run groups are listed [here](#) and
261 documented in Ref. [3]. The calibration outside of these runs is only approximate and is currently
262 not supported by the calorimeter calibrations group.

263 7 Systematics

264 The calibration systematics are separated into two categories. The first of which are systematic
265 errors/uncertainties assuming HIJING and GEANT4 perfectly model the truth events and the
266 detector response, respectively. These include,

- 267 • ϕ symmetry assumption in the TSC
- 268 • Statistical uncertainty of the TSC
- 269 • Statistical precision on the η -by- η π^0 mass
- 270 • Various analysis choices: E_T cuts on clusters and towers
- 271 • Uncertainties on vertex location

272 The second category includes mis-modeling by the MC and includes,

- 273 • Additional smearing in data leading to shifts in the π^0 mass
- 274 • effects from masked towers on π^0 mass
- 275 • effects from larger noise in data

276 All of these items effect the EMCal in different ways, stochastic vs. systematic, tower-bytower
277 vs. η -by η , etc. Because the systematics are delivered to the user by manipulating the tower-
278 level calibrations and having user reruns their analysis, the above systematics are grouped into
279 categories of topology and type of systematic effect. The systematic effect arise from discrepancies
280 with MC and come from specific calculations. The stochastic ones are just randomly generated
281 adjustments to the calibration according to some calibration uncertainty arising from limited
282 statistics. Given the current set of systematics the categories are:

- 283 1. Tower-by-tower (TBT) statistical effect
- 284 2. tower-by-tower systematic effect
- 285 3. η -by- η (EBE) statistical effects

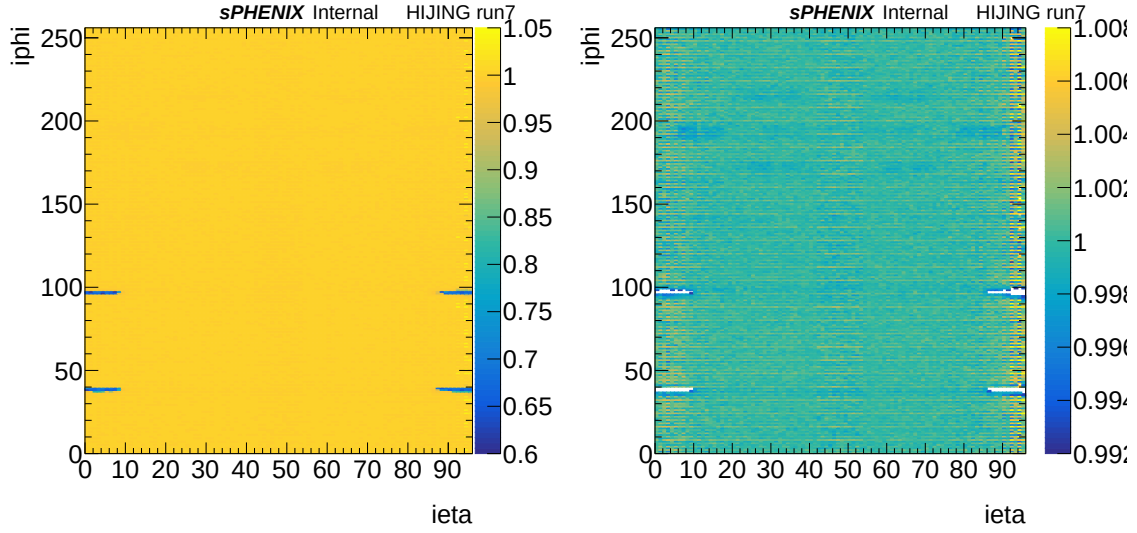


Figure 17: The TSC factors found in run 7 HIJING where the right panel is a zoomed in version of the left panel.

4. overall scale effects (*scale*)

Table 2 summarizes the list of systematics and their characteristics.

Description	Topology	Type	Size est.
ϕ symmetry assumption in the TSC	TBT	syst	< 1%
Stats of the TSC	TBT	stochastic	2%
Stats of η -by- η π^0 mass	EBE	stochastic	3%
vertex	EBE	syst	< 1%
Analysis choices			
Data/MC noise	scale	syst	1%
Data/MC smearing	scale	syst	1%

Table 2: Table of EMCal calibration systematic uncertainties.

7.1 TSC related uncertainties

The first uncertainty associated with the TSC is the assumption of ϕ symmetry of EMCal towers and the upstream material. This is only approximately true. The error incurred by this assumption is probed in HIJING where the calibration factor is by definition 1, because we are calibrating to the MC. Thus any non-unity calibration generated by the TSC in MC is an uncorrected issue. Fig. 17 is the TSC output from HIJING run 7. These factors are delivered to the user in systematic nodes TOWERINFO_CALIB_SYST3UCEMC and TOWERINFO_CALIB_SYST3DCEMC, where one is the application of the TSC factors and the other is the inverse factors.

The second uncertainty is associated with the statistical limitation of the TSC. Fig. 18 is a 1D distribution of the TSC factors in the second iteration of the method. In the ideal case these

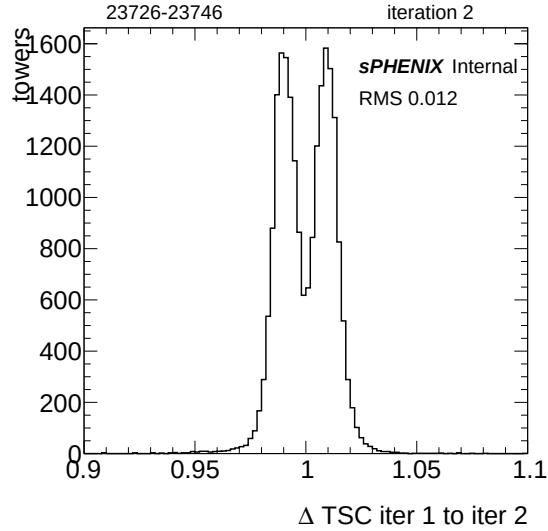


Figure 18: A 1D distribution of TSC factors for the second iteration where the scatter is interpreted as statistical-related non-convergence.

would all be 1 given the first iteration should remove all calibration issues. The scatter is greater than what is expected from detector geometry alone and is attributed to statistical scatter. A 2% stochastic TBT uncertainty is delivered to the user in systematic node TOWERINFO_CALIB_SYST4CEMC. This systematic should be symmetrized in final results unless it is clearly demonstrated that the smearing effect pushes results in one direction.

7.2 π^0 mass calibration errors

The first uncertainty in the π^0 calibration is incurred by the large uncertainties on the data π^0 mass. These uncertainties can be seen in Fig. 10 and is about 3 MeV and is rounded to a 2% stochastic uncertainty on the EBE calibration and is passed to the user via node TOWERINFO_CALIB_SYST1CEMC and should be symmetrized unless this smearing clearly pushes the results in one direction.

7.3 Data/MC discrepancies related uncertainties

The diverse set of data/MC differences and their downstream effects on the calibration are still an active work in progress. The first step towards this is the uncertainty in the data resolution compared to MC. This is documented in Table 1, where the added level of smearing is varied by 1% and the observed difference in the resulting predicted π^0 mass for data is $\approx 1\%$ up or down. There are also studies of the input tower energy lower bound having different effects in data and MC, showing some discrepancy of the 1% level. Currently the overall scale systematic uncertainty is set to 2% and is about the level that is expected at the end of these studies. It is delivered to the user in the TOWERINFO_CALIB_SYST2CEMC node and should be symmetrized.

References

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